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Tumour Regression of Uveal Melanoma after Ruthenium-106 Brachytherapy or Stereotactic Radiotherapy with Gamma Knife or Linear Accelerator

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Key Words

Uveal melanoma · Radiotherapy · Ruthenium · Gamma knife · Linear-accelerator-based stereotactic teletherapy · Ultrasound

Abstract

Purpose: This study assesses differences in relative tumour regression and internal acoustic reflectivity after 3 methods of radiotherapy for uveal melanoma: (1) brachytherapy with ruthenium-106 radioactive plaques (RU), (2) fractionated high-dose gamma knife stereotactic irradiation in 2–3 fractions (GK) or (3) fractionated linear-accelerator-based stereotactic teletherapy in 5 fractions (Linac). ***Methods:*** Ultrasound measurements of tumour thickness and internal reflectivity were performed with standardised A scan pre-operatively and 3, 6, 9, 12, 18, 24 and 36 months postoperatively. Of 211 patients included in the study, 111 had a complete 3-year follow-up (RU: 41, GK: 37, Linac: 33). Differences in tumour thickness and internal reflectivity were assessed with analysis of variance, and post hoc multiple comparisons were calculated with Tukey's honestly significant difference test. ***Results:*** Local tumour control was excellent with all 3 methods (>93%). At 36 months, relative tumour height reduction was 69, 50 and 30% after RU, GK and Linac, respectively. In all 3 treatment groups, internal reflectivity increased from about 30% initially to 60–70% 3 years after treatment. ***Conclusion:*** Brachytherapy with ruthenium-106 plaques results in a faster tumour regression as compared to teletherapy with gamma knife or Linac. Internal reflectivity increases comparably in all 3 groups. Besides tumour growth arrest, increasing internal reflectivity is considered as an important factor indicating successful treatment.

Introduction

Uveal melanoma is the most common intraocular malignancy with an incidence of 6 cases per million population per year [1]. Diagnosis is based on clinical criteria and ultrasound findings [2]. Ultrasound B scan is used to localise the tumour and to measure tumour base dimensions. We use standardised A scan to get very precise and reliable measurements of tumour thickness and internal acoustic reflectivity.

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Treatment options depend on tumour size and localisation. Malignant melanoma of the uvea may be treated by transpupillary thermotherapy [3, 4], brachytherapy (ruthenium-106 [5]; iodine-125 [6], palladium-103 [7]), charged-particle teletherapy [6, 8, 9], surgical techniques [10] and combinations of surgery and brachytherapy. Within the last few years, new and interesting applications of modern stereotactic external photon beam teletherapy have been introduced in ocular oncology. Clinical studies using the multi-cobalt-60 device Leksell Gamma Knife® or modified linear accelerators have been initiated to study their impact for treating uveal melanoma [11–13].

Success of all radiotherapeutic procedures can be defined in many ways: first of all in survival, then staying free from metastases, preserving the eye, staying free from pain or side-effects, preserving vision or achieving local tumour control. Regarding metastasis rates and survival of patients, all local treatment options are similarly effective [14–17]. Regarding the other outcome parameters, the optimal local treatment option is still a matter of discussion. Because some treatment options are limited to special indications, direct comparison is difficult. Prospective, dynamically balanced trials have been rarely performed [6].

In radiotherapy, local tumour control (defined as an arrest of tumour growth with or without tumour shrinkage) measured by ultrasound or other imaging techniques indicates successful treatment. In our institution, we use standardised A scan for monitoring tumour regression. Since the fundamental work of Ossoinig [18] many studies have been performed to evaluate the echographic response of uveal melanomas to radiation treatment. After successful brachytherapy with ruthenium-106 the tumour usually responds well with often only a flat scar remaining [5, 19]. As the tumour decreases in size, the internal acoustic reflectivity increases. Increasing internal reflectivity represents morphological changes within the tumour stroma indicating treatment success. It represents replacement of neoplastic with necrotic, fibrous or cicatricial tissue [2, 20, 21].

This study assesses relative tumour regression and changes in internal acoustic reflectivity after 3 different methods of radiotherapy for uveal melanoma: (1) ruthenium-106 radioactive plaque brachytherapy (RU), (2) fractionated high-dose gamma knife stereotactic teletherapy (GK) and (3) fractionated linear accelerator-based stereotactic teletherapy (Linac). Of course, total dose, dose rate and dose volumes and also the tumour localisation differ between the 3 methods. This is a major drawback of this study, but it is the only one with standardised follow-up investigations performed in a single centre for 3 different radiotherapy methods.

Methods

For this retrospective study, we identified 74 patients with RU, 58 patients with GK and 79 patients with Linac with a minimal follow-up time of 12 months and complete ultrasound evaluations in our out-patient oncology service. Patients were not randomised to one or another treatment but were managed selectively depending on the clinical findings.

Brachytherapy with episcleral application of ruthenium-106 radioactive plaques is used in our department to treat uveal melanomas located in the middle or peripheral choroid (>3 mm distance of central tumour margin to optic disc or macula) and with a thickness between 3 and 7 mm [22]. We aim to deliver a minimal apical tumour dose of 100–120 Gy and a maximal scleral dose of 1,000 Gy which is standard for ruthenium-106 radioactive plaques [23].

Between 1993 and 1997, we performed a prospective study on fractionated high-dose gamma knife stereotactic radiotherapy (Gamma Knife®, Elekta, Sweden) [13]. This treatment option was offered as an alternative treatment to enucleation only to patients with large tumours in the beginning. Inclusion criteria were (1) melanomas of >7 mm initial height or (2) juxtapapillary and/or juxtamacular tumours (height >3 mm, less than 3 mm distance to optic disc or macula). During the last years of this study also smaller tumours have been treated with this method. The total radiation dose varied between a total dose of 70 Gy (delivered in 2 fractions of 35 Gy) and a total dose of 45 Gy (delivered in 3 fractions of 15 Gy). The planning target volume was covered with the 50% isodose line.

In 1997 we started a prospective study with an increased number of fractions using Linac (6MV Linac, Saturne 43®, General Electric Medical System, Paris, France). At the start, the treatment inclusion criteria were the same as mentioned above for the GK group. We used a total dose of 70 Gy (5 fractions) for the first 24 consecutive patients, then the total dose was lowered to 60 Gy (5 fractions). The 80% isodose covered the planning target volume. Patients initially presenting with extrascleral extensions or metastatic disease did not enter the study protocol. Further details of the two latter methods have been published [12, 13, 24].

Between the 3 groups, we compared tumour thickness reduction measured pre-operatively and 3, 6, 9, 12, 18, 24 and 36 months post-operatively with standardised A scan. If the thickness was reduced to an amount not measurable with the A scan and in the biomicroscopy examination a scar formation was seen, the outcome was called 'flat scar'. Internal tumour reflectivity was calculated by the device CineScanS (Quantel Medical). Local tumour control was defined as tumour growth arrest or regression.

Statistical Analysis

The effect of the 3 different treatment options on relative tumour thickness and internal reflectivity up to 3 years was tested with Anova (analysis of variance). Post hoc multiple comparisons were calculated with Tukey's honestly significant difference test to analyse differences in between the 3 groups. An α level of 0.05 was considered as statistically significant.

Table 1. Patient, treatment and tumour characteristics: means ± standard deviation

	RU	GK	Linac
Number of patients at baseline	74	58	79
Minimum follow-up 1 year	73	54	78
Minimum follow-up 2 years	52	43	45
Minimum follow-up 3 years	41	37	33
Patient age, years	58 ± 15 (23–88)	60 ± 13 (37–88)	59 ± 14 (21–89)
Follow-up, months	55 ± 36 (12–202)	47 ± 18 (11–86)	29 ± 12 (11–54)
Scleral dose, Gy	970 ± 287 (331–1571)		
Apical dose, Gy	136 ± 44 (50–280)		
Marginal dose, Gy		58 ± 13 (45–70)	63 ± 5 (60–70)
Height of tumour, mm	5.0 ± 1.5 (2.4–8.5)	8.4 ± 2.6 (3.3–14.3)	5.2 ± 1.9 (2.7–10.4)
Largest diameter, mm	10.9 ± 3.7 (3.0–20.0)	15.0 ± 3.7 (7.7–25.0)	10.9 ± 2.6 (6.8–18.1)
Tumour volume, mm^3	311 ± 233 (41–1021)	985 ± 784 (102–3387)	317 ± 264 (64–1461)
Distance to optic disc, mm	5.4 ± 3.0 (0–15)	2.8 ± 2.8 (0–11)	2.4 ± 1.8 (0–8)
Distance to fovea, mm	4.7 ± 3.5 (0–15)	2.6 ± 2.7 (0–12)	1.6 ± 1.5 (0–6)
Anatomical location			
Choroid	65	42	73
Choroid and ciliary body	4	1	0
Ciliary body	5	15	1

Figures in parentheses indicate minimum and maximum values.

Results

Patient, treatment and tumour characteristics are summarised in table 1. Of initially 74 patients with RU treatment, 41 were available for sonographic follow-up after 3 years (excluding 4 enucleations, 4 deaths, 15 missed follow-up, 5 lost to follow-up, 5 time censoring). In the GK group of initially 58 patients, 37 had a 3-year sonographic follow-up (9 enucleations, 9 deaths, 3 lost to follow-up). Of the 79 Linac patients, 33 had a 3-year follow-up (5 enucleations, 10 lost to follow-up, 31 time censoring). In the RU group, the distance to the optic disc or fovea was significantly larger, resulting from the inclusion criteria mentioned above. The GK group included significantly more ciliary body tumours.

With RU, a mean dose of 977 Gy to the sclera and a mean dose of 137 Gy at the tumour apex were achieved with a dose rate of 10–15 cGy/min and an application time of 4–10 days. With GK, a mean marginal dose of 58 Gy (resulting in a mean centre dose of 114 Gy) was reached with a dose rate of 2–4 Gy/min at the centre, delivered in 2–3 fractions. Similar radiation values were achieved with Linac (63 Gy mean marginal dose delivered in 5 fractions).

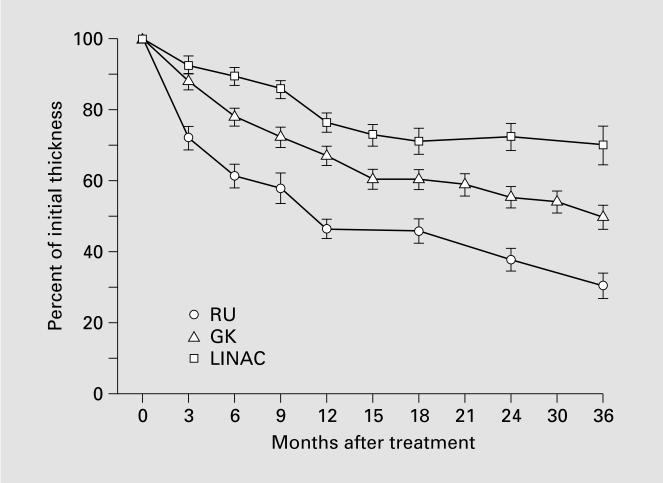
The initial tumour thickness was found to be significantly higher in the GK group as compared to the 2 other groups. Therefore, the relative tumour reduction is shown for the 3 groups in figure 1 for better comparison. At 12 months, tumour height reduction was 54, 33 and 23%, at 24 months 62, 44 and 28% and at 36 months 69, 50 and 30%, after RU, GK and Linac, respectively.

Anova revealed a significant difference in tumour thickness reduction after the 3 treatment options. The post hoc analysis showed significant differences ($p < 0.01$) in between all the 3 groups in tumor thickness reduction.

The development of internal reflectivity after treatment is shown in figure 2. For all 3 groups, internal reflectivity increased from initially 30% to 60–70% after 3 years. In the RU group, the reflectivity increase was slightly pronounced. In contrast to tumour thickness reduction (fig. 1), no statistically significant differences between the 3 groups were found concerning internal reflectivity (Anova).

A very high local tumour control rate of 97 and 98% after 1 and 2 years, respectively, was accomplished with RU brachytherapy. A flat scar formation was seen in 70% after 3 years.

In the GK group, local tumour control rates were 96 and 100% after 1 and 2 years, respectively. In the Linac group, the local tumour control rates were 93 and 94% after 1 and 2 years, respectively. After 3 years, a flat scar remained only in 15% after GK and Linac; in 85%, a residual tumour was observed in these groups.



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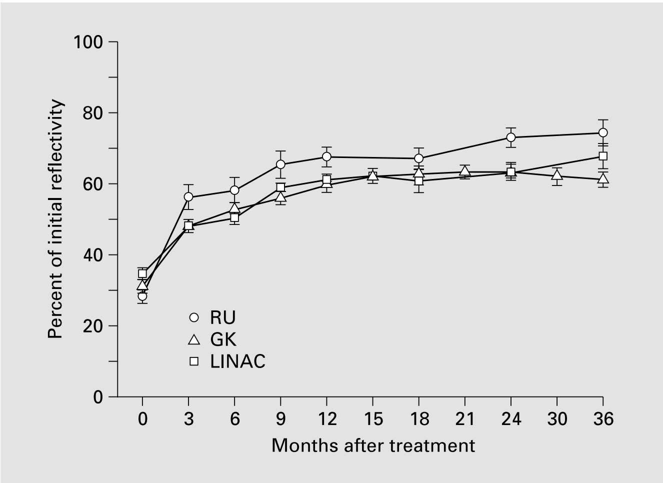

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Fig. 1. Reduction of relative tumour thickness; means ± SEM.
Fig. 2. Development of internal reflectivity (A scan); means ± SEM.

Discussion

This study compared data on tumour regression after 3 different methods of radiotherapy for uveal melanoma collected at a single institution. Furthermore, this study includes the largest series of consecutive GK and Linac patients with a regular follow-up. Drawbacks of our study are (1) the non-randomised manner of patient assignment to treatment, and (2) in the gamma knife group also large tumours of ciliary body origin were included, which are known to have a poorer local and systemic prognosis [25]. Tumour regression depends not only on treatment, but also on the histological type of the uveal melanoma [26], which remains unknown in our patients.

Standardised A scan is a very precise method for exactly monitoring tumour regression [27]. The mean interobserver variation of A scan tumour thickness measurement is 0.7 mm when comparing observers from different institutions, and less than 0.5 mm for experienced observers [27].

Ravozzoni et al. [28] compared tumour regression after brachytherapy with ruthenium-106 (n = 12) and teletherapy with accelerated protons (n = 20). Comparably to our results, a significantly greater tumour reduction after brachytherapy was found [28].

After charged-particle radiotherapy, such as protons or helium ions, a very high local tumour control rate of 96% is achieved [8, 9, 29].

Another study group has evaluated the echographic behaviour of 29 uveal melanomas treated radiosurgically with a gamma knife unit. In a group of 20 patients with an initial thickness of 5 mm or more a residual prominence was found in 75%. The increase in internal reflectivity is also emphasised as a good marker for effective treatment [20].

We found a fast and more pronounced local tumour regression in the RU group, as compared to the 2 teletherapy groups. After 3 years, in 70% a flat scar formation was observed in the RU group. In contrast, only in 15% was a flat scar seen after 3 years in the GK and Linac groups.

The differences between the groups must be related to the different radiation characteristics. During brachytherapy there is a large gradient between the tumour base and tumour apex whereas in fractionated teletherapy the radiation dose is delivered in a more uniform distribution [30]. With brachytherapy, the tumour base receives an extremely high irradiation dose, because the radiation dose rapidly decreases with distance to the radioactive source and the tumour apex has to be treated sufficiently. The uvea and the tumour feeding vessels are localized within this field of high irradiation doses. These 2 factors obviously contribute to the faster and more complete tumour regression seen after brachytherapy.

In teletherapy, the tumour and the surrounding tissue are irradiated with relatively homogeneous radiation fields. In contrast to brachytherapy, the tumour base does not receive a substantially different dose. This might explain why in both teletherapy groups (GK, Linac) the tumour shrinkage was less pronounced than in the brachytherapy group (RU). With GK, where the central part of the tumour receives the double of the peripheral dose, the shrinkage was still more pronounced in comparison to the Linac-treated tumours. In this group, the tumors were irradiated with an almost uniform irradiation.

Despite the differences in local tumour height and volume regression, the increase in internal reflectivity was surprisingly similar in all 3 groups. This might indicate that treatment was effective also in the groups with slower local tumour regression (GK, Linac). Increasing internal reflectivity represents treatment success even without marked tumour regression. Therefore, we should focus not only on quantitative amounts of tumour regression but also on qualitative changes in the tumour stroma represented by internal reflectivity [2, 20, 21] when judging treatment success after radiotherapy for uveal melanoma. Thus, in patients with increasing internal reflectivity we are less concerned about the lack of tumour regression.

In conclusion, this study shows that brachytherapy with ruthenium-106 plaques results in faster tumour regression within the first 3 years compared to teletherapy with gamma knife or Linac. After 3 years, in 70% a flat scar remains after RU, whereas after GK or Linac a flat scar remains in only 15%; in 85%, a residue persists after teletherapy. The investigated eye-preserving treatment options for uveal melanoma have shown different regression characteristics but an equivalent rise in internal reflectivity, indicating successful treatment independent of the amount of tumour shrinkage.

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